

Project Title: Linux Hardening Audit Tool v1.0

1. ABSTRACT

The Linux Hardening Audit Tool v1.0 is a Python-based command-line security auditing tool developed to evaluate the security posture of a Linux system. The project focuses on implementing automated hardening checks aligned with fundamental Linux security best practices. The tool performs multiple system-level checks, calculates an overall compliance score, and generates structured audit reports in TXT and JSON formats. The primary objective of this project was to gain hands-on experience in Linux system security auditing, automation using Python, and modular software design.

2. INTRODUCTION

Linux systems are widely used in enterprise environments, cloud infrastructure, and cybersecurity domains. Proper system hardening is essential to reduce attack surfaces and prevent unauthorized access. Manual security verification is time-consuming and error-prone. Therefore, automated auditing tools are essential for assessing system configurations efficiently.

The Linux Hardening Audit Tool v1.0 was developed to automate essential security checks, evaluate compliance, and provide structured output for analysis. This project demonstrates practical implementation of cybersecurity principles using Python scripting and Linux system commands.

3. TOOLS AND TECHNOLOGIES USED

- Programming Language: Python 3
- Operating System: Kali Linux (Virtual Machine)
- Virtualization Platform: Oracle VirtualBox
- Version Control: Git and GitHub
- Execution Mode: Command Line Interface (CLI)
- Required Privileges: sudo (for system-level checks)

Python was used to automate Linux command execution and process system configuration data. The tool was tested inside a Kali Linux virtual machine environment to simulate real-world security assessment conditions.

4. STEPS INVOLVED IN BUILDING THE PROJECT

Step 1: Requirement Identification

Identified core Linux hardening checks such as firewall status, SSH configuration, password policy, UID validation, and file permission analysis.

Step 2: Modular Design

The project was structured into three main modules:

- main.py (execution controller)
- audit_checks.py (security check implementations)
- report_generator.py (report creation module)

This modular design allows easy scalability and addition of new security checks.

Step 3: Implementation of Security Checks

Implemented system-level checks including:

- Firewall (UFW) status verification
- SSH root login configuration check
- Password aging policy validation
- UID 0 account verification
- SUID file detection
- World-writable file detection
- Listening services detection

Step 4: Compliance Score Calculation

Critical checks were assigned scoring logic to calculate overall compliance percentage. Based on the score, the system is categorized as:

- SECURE
- MODERATE RISK
- HIGH RISK

Step 5: Report Generation

Developed functionality to generate:

- audit_report.txt (human-readable report)
- audit_report.json (structured report format)

Step 6: Testing and Validation

The tool was executed using sudo privileges to ensure system-level access. Output was validated and optimized for clean terminal display.

5. CONCLUSION

The Linux Hardening Audit Tool v1.0 successfully implements automated Linux system security checks using Python. The project demonstrates understanding of Linux hardening concepts, compliance evaluation, and modular tool development.

This project enhanced practical knowledge of:

- Linux security configurations
- Command-line automation
- Compliance scoring mechanisms
- Structured report generation
- Version control and release management

The tool is extensible and can be further improved by integrating CIS benchmarks, logging mechanisms, and advanced risk mapping. The project reflects foundational cybersecurity engineering skills relevant to roles such as Security Analyst, SOC Analyst, and Junior Security Engineer.